

LECTURE 1 — PERITONEUM, MESENTERY & RETROPERITONEUM

Peritoneum — anatomy & physiology

- **Largest cavity** in body (~2 m² lining ≈ skin area). Mesothelium on fibroelastic tissue + lymphatic/capillary plexus. Only a few mL fluid normally (pale yellow, lymphocytes).
- **Parietal peritoneum: richly innervated** → **severe, accurately localized (somatic) pain**. **Visceral: poorly innervated** → **poorly localized midline pain**.
- **Functions:** health — lubrication, fluid/particulate absorption; disease — pain perception, inflammatory/immune response, fibrinolysis.
- Fluid circulates upward to **diaphragm** (absorbed via diaphragmatic pores). Abscesses collect at **pelvis + subdiaphragmatic** spaces (gravity, standing vs lying).

Peritonitis

- Inflammation of peritoneum: localized/diffuse, acute/chronic.
- **Causes:** bacterial (GI/non-GI), chemical (bile, barium), allergic (starch), traumatic, ischaemic (strangulated bowel), misc (FMF).
- **Paths to infection:** GI perforation (ulcer, appendix, diverticulum), transmural translocation (pancreatitis, ischaemia, primary), exogenous (drains, dialysis), female genital tract (PID), haematogenous (rare).
- **Localized peritonitis:** somatic pain, **involuntary guarding + rebound tenderness = peritonism**. Diaphragm → **shoulder tip (phrenic, C5) referred pain**. Pelvic → signs on PR/PV exam.
- **Diffuse peritonitis** = life-threatening (perforated viscus/anastomotic leak).
- **Clinical:** pain worse on movement/cough, **Hippocratic facies**, **lies still**, anorexia/fever, N/V, ± pyrexia, tachycardia, **guarding/rigidity (board-like)/rebound**, absent bowel sounds, **septic shock (SIRS/MODS)** late. Signs masked in obese/immunosuppressed.
- **Investigations:** **CT = investigation of choice**; erect CXR (subdiaphragmatic gas); laparoscopy if inconclusive.
- **Management:** general (fluids/electrolytes, NG tube, urinary catheter, **broad-spectrum antibiotics**, analgesia, support) + **surgical "source control"** + peritoneal lavage/drainage.

Specific peritonitis types

- **Acute bacterial:** mostly GI perforation; primary "spontaneous" (strep/pneumococcal/Haemophilus). Non-GI = pelvic (Chlamydia, gonococci); **Fitz-Hugh-Curtis** (perihepatitis, Glisson's capsule scarring).
- **Biliary:** post-cholecystectomy (clip slippage, accessory duct, CBD perforation). Localized → drain + ERCP stent; diffuse → surgery.
- **Spontaneous bacterial peritonitis (SBP):** ascitic fluid infection; cirrhosis/ascites. **Diagnosis** = **paracentesis**, **neutrophils >250/mm³**; culture negative in 60%; usually E. coli/strep. **Empirical 3rd-gen cephalosporin (cefotaxime)** immediately.
- **Tuberculous:** common in poor countries, rising with HIV. Ileocaecal/mesenteric nodes. **Wet (ascitic) type 90%** vs dry/plastic (matted loops → subacute obstruction). Ascites = exudate (protein >25–30, lymphocytes >40%); **adenosine deaminase high sens/spec**; smear/culture slow (4–8 wks). Treat anti-TB (± surgery for obstruction).

Peritoneal tumours

- **Primary (mesothelioma)**: rare, asbestos-linked, lethal → cytoreductive surgery + **HIPEC** or cisplatin chemo.
- **Secondary (peritoneal carcinomatosis)**: common; ovarian/GI malignancy; **omental cake**; CT/MRI + histology (distinguish from TB). Resectable → peritonectomy + **HIPEC** (heated 41–42°C chemo, 90 min; standard for **pseudomyxoma peritonei**).

Intraperitoneal collections

- **Ascites**: excess production/reduced absorption; clinically apparent **>1.5 L**; flank dullness, **shifting dullness**, fluid thrill.
- **Transudate (<25 g/L protein)**: low albumin, malnutrition, nephrotic, protein-losing enteropathy, CCF, **portal hypertension (cirrhosis)**.
- **Exudate (>25 g/L)**: peritoneal malignancy, TB, **Budd-Chiari**, pancreatic, chylous, **Meigs' syndrome**.
- Treat cause; **TIPS** if portal HTN; sodium restriction + **spironolactone + furosemide**; therapeutic paracentesis (large-volume + colloid).
- **Chylous ascites**: milky (chylomicrons); mostly **lymphoma**; treat fat-free diet + MCT.
- **Intraperitoneal abscess**: pus collection (subphrenic, pelvic, paracolic, subhepatic). Symptoms: malaise, failure to recover post-op, **swinging pyrexia**, shoulder tip (subphrenic). **CT = diagnosis + drainage guide**. **<5cm → IV antibiotics**; **>5cm → percutaneous/surgical drainage**.

Mesentery

- **Mesenteric adenitis**: inflamed mesenteric nodes (ileocaecal); viral/Yersinia/Campylobacter/TB, post-URTI/SARS-CoV-2. RIF somatic pain, **marked fever**, **cervical nodes**; **shifting tenderness differentiates from appendicitis**. Leukocytosis day 1 (falls day 2). Supportive.
- **Acute mesenteric ischaemia**: 3 vessels (coeliac, SMA, IMA). Usually **embolism to coeliac/SMA**. **Pain out of proportion to findings**; necrosis → peritonism. **Marginal artery of Drummond** protects sigmoid/descending colon after IMA division (AAA repair).
- **Chronic mesenteric ischaemia**: atherosclerotic origin narrowing → **postprandial pain + fear of eating + weight loss**; CT angiography; stent/endarterectomy.
- **Mesenteric cysts**: small bowel mesentery (60%). **Chylolymphatic** (commonest, lymph-filled, independent blood supply → enucleate); **enterogenous** (mucinous, shared blood supply → resect bowel).
- **Mesenteric tumours**: benign (lipoma, desmoid); malignant (lymphoma, secondary carcinoma, NET).

Retroperitoneum

- **Psoas abscess**: historically **TB spine (Pott's)**; now mostly **Crohn's** or spread from GI/urinary tract. Back pain, fever, **groin swelling (tracks under inguinal ligament)**, pain on hip extension/fixed flexion. CT → **percutaneous drainage + antibiotics**.
- **Retroperitoneal lipoma**: large; **myxomatous degeneration** (unique to retroperitoneum); rapid growth → liposarcoma.
- **Retroperitoneal sarcoma**: rare (1–2% solid malignancies); 5th decade; **liposarcoma**, **leiomyosarcoma**, **malignant fibrous histiocytoma**. Present **late + large** (non-specific symptoms). CT/MRI; treat **surgical resection** (chemo/RT poor); survival 35–50%.

Lecture 1 — high-yield

- **Parietal** = localized somatic pain; **visceral** = poorly localized midline.

- **Peritonism = guarding + rebound; diaphragm irritation → shoulder tip (C5).**
- **Peritonitis: CT = investigation of choice; treat source control + lavage + antibiotics.**
- **SBP: neutrophils $>250/\text{mm}^3$, cefotaxime; ascites transudate <25 vs exudate >25 g/L.**
- **Peritoneal carcinomatosis = omental cake; HIPEC for pseudomyxoma.**
- **Mesenteric adenitis: shifting tenderness vs appendicitis. Acute ischaemia: pain out of proportion.**
- **Psoas abscess = Crohn's (or TB spine); CT drainage.**

LECTURE 2 — INTESTINAL OBSTRUCTION

Classification

- **Dynamic (mechanical):** peristalsis against a mechanical block.
- **Adynamic:** no mechanical block; peristalsis absent/inadequate (**paralytic ileus, pseudo-obstruction**).
- **By site:**
- **Intraluminal:** faecal impaction, foreign body, **bezoars, gallstone ileus**.
- **Intramural:** stricture (Crohn's, TB), **malignancy**, intussusception, volvulus.
- **Extramural:** **bands/adhesions, hernia**.

Causes

- **Bolus:** gallstones, food (post-gastrectomy), **trichobezoar (hair)/phytobezoar (fibre)**, stercoliths, worms (Ascaris).
- **Adhesions** (commonest SBO): post inflammation/anastomosis, foreign material, infection, Crohn's, radiation. **Prevention:** good technique, saline washout, minimize gauze contact, cover raw surfaces.
- **Bands:** congenital (vitellointestinal duct), post-surgery, omental.

Pathophysiology

- **Proximal:** dilates; peristalsis $\uparrow$ then fails $\rightarrow$ flaccid/paralysed. **Distal:** normal until empty $\rightarrow$ collapses.
- **Distension:** gas (mostly **nitrogen 90%** + **H₂S** from bacterial overgrowth) + fluid (saliva/gastric/bile/pancreatic, up to 2.5 L/24h, retained).
- **Systemic:** dehydration + electrolyte loss ($\downarrow$ intake, $\downarrow$ absorption, vomiting, **third-spacing**, transudation). $\rightarrow$ **oliguria, dry tongue, $\uparrow$ urea + haematocrit**. **Hypokalaemia NOT a feature of simple obstruction.**

Strangulation (emergency)

- Blood supply compromised $\rightarrow$ ischaemia/infarction. Mechanisms: **direct pressure (hernia, bands), interrupted mesenteric flow (volvulus, intussusception), $\uparrow$ intraluminal pressure (closed loop)**.
- **Venous return compromised before arterial** $\rightarrow$ haemorrhagic infarction $\rightarrow$ translocation/endotoxin.

Closed loop

- Distension confined to loop (proximal not marked). **High risk of strangulation/perforation.** Classic: **malignant colonic stricture + competent ileocaecal valve**.

Intussusception & volvulus

- **Intussusception:** gut invaginates into adjacent segment. **Children 5–10 months** (idiopathic); **adults = pathological lead point (polyp/tumour)**. **Ileocolic commonest**. CT target/sausage sign.
- **Volvulus:** axial twist on mesentery. Obstruction $>180^\circ$, vascular occlusion $>360^\circ$. **Sigmoid commonest in adults**; may decompress per anum.

Clinical features — the cardinal quartet

- **Pain:** sudden, severe, colicky — umbilical (SBO) / lower abdomen (LBO); becomes constant/diffuse with distension.
- **Vomiting:** high SBO early + profuse (minimal distension); low SBO later + **faeculent**; LBO late.
- **Distension:** minimal in high SBO; **early/pronounced in LBO**.
- **Absolute constipation** (no faeces or flatus) = cardinal of complete obstruction. **Exceptions:** Richter's hernia, gallstone ileus, partial/subacute (may have diarrhoea).
- **Strangulation signs:** constant severe pain (**not controlled by opiates**), tenderness + rigidity + peritonism, **shock**, pyrexia.

Imaging

- **Plain AXR:** SBO = central, transverse, **valvulae conniventes** (cross full width, "ladder"), little colonic gas; LBO = peripheral, **haustra** (don't cross fully), **caecal gas**. Erect → fluid levels.
- **CT** highly accurate; ischaemia hard (reduced wall enhancement; absent mesenteric fluid ↓probability); **strangulation diagnosis remains clinical**. Intussusception = **target/sausage sign**.
- **Water-soluble contrast** (SBO no strangulation): reaches colon 4–24h → resolution 96% likely; fails → surgery ~90%. **Barium contraindicated in acute obstruction**.

Management

- **Supportive (resuscitate before surgery unless strangulation/closed loop):** NG tube (decompress + ↓aspiration), fluid/electrolytes (Hartmann's/saline).
- **Early surgery indications:** obstructed external hernia, strangulation signs, obstruction in **previously unoperated abdomen**. Adhesive obstruction → conservative up to **72h** if no ischaemia.
- **Surgical principles:** manage obstruction site + proximal bowel + cause. Intraop decompression (milk retrograde to orogastric tube). **Viability** = colour, sheen, peristalsis, mesenteric pulsation after **hot packs 10 min**; uncertain → resect. (Paul-Mikulicz for sigmoid volvulus.)

Lecture 2 — high-yield

- **Dynamic quartet:** pain, vomiting, distension, absolute constipation. SBO central/early vomit; LBO peripheral/late vomit/early distension.
- **Adhesions = commonest SBO; hernia next.** LBO commonest = colorectal cancer.
- **Strangulation = constant pain (not opiate-controlled) + peritonism + shock = emergency.** Venous before arterial.
- **Closed loop = competent ileocaecal valve + colonic obstruction → perforation risk.**
- **SBO = valvulae conniventes (ladder); LBO = haustra.** Hypokalaemia NOT in simple obstruction.
- **Water-soluble contrast predicts SBO resolution; barium contraindicated.**
- **Resuscitate first (NG + fluids) UNLESS strangulation/closed loop → immediate surgery.**